

NEW DELHI INDIRA GANDHI INTL, India

WMO: 421810

Lat: 28.567N

Lon: 77.103E

Elev: 777

StdP: 14.29

Time Zone: 5.50 (IND)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	42.6	44.5	31.2	26.1	69.8	34.1	29.6	72.3	16.6	65.2	14.5	64.2	3.4	240	0.409

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	22.7	110.8	72.0	107.7	72.1	105.5	72.3	84.6	91.8	83.6	90.7	82.6	89.3	10.3	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
83.4	179.8	87.5	82.3	173.0	87.0	80.8	165.0	86.2	49.8	90.6	48.4	91.1	47.4	89.2	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.0	14.3	12.8	DB	37.9	114.5	2.8	2.8	35.8	116.6	34.2	118.2	32.6	119.8	30.6	121.8	(4)
			WB	37.1	86.2	2.7	2.1	35.1	87.8	33.5	89.0	32.0	90.2	30.0	91.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	78.3	56.8	63.9	74.1	86.5	92.9	88.6	87.1	85.8	80.4	70.3	60.1		
		DBStd	13.24	4.69	5.35	6.26	6.00	5.33	5.58	4.01	3.03	3.28	4.05	4.36	4.82	(7)
		HDD50	5	4	0	0	0	0	0	0	0	0	0	1		
		HDD65	512	258	79	5	0	0	0	0	0	0	7	163		
		CDD50	10348	215	388	746	1095	1330	1288	1198	1149	1073	943	610	313	(10)
		CDD65	5379	4	46	286	645	865	838	733	684	623	478	167	10	(11)
		CDH74	75802	43	447	3252	9619	14117	13470	10434	9299	8149	5386	1457	129	(12)
		CDH80	44968	5	97	1505	6095	9825	9221	6031	4917	4191	2656	420	5	(13)
(14)	Wind	WSAvg	5.5	4.8	5.6	5.9	6.4	7.2	7.4	6.6	6.0	5.3	3.8	3.4	3.8	(14)
(15)	Precipitation	PrecAvg	23.0	0.4	0.9	0.4	0.4	1.1	2.1	6.3	6.3	4.3	0.4	0.2	0.2	(15)
		PrecMax	38.7	2.3	4.3	3.5	2.5	5.9	6.1	23.1	17.8	11.9	2.9	1.6	2.0	(16)
		PrecMin	9.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
		PrecStd	7.2	0.6	1.1	0.8	0.6	1.5	1.8	5.3	4.2	3.4	0.8	0.4	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.9	87.7	100.4	109.6	113.4	114.5	105.7	100.5	100.4	98.5	91.2	80.6	(19)
			MCWB	61.7	64.9	66.4	69.9	71.1	75.3	78.1	76.2	75.5	69.9	66.0	60.6	(20)
		2%	DB	73.6	82.6	95.3	106.2	111.1	110.8	102.3	98.4	98.3	95.4	87.5	77.3	(21)
			MCWB	58.7	63.0	66.4	68.7	70.7	74.1	78.8	78.1	76.1	71.0	65.1	59.7	(22)
		5%	DB	70.2	79.1	91.6	104.0	108.3	107.6	99.0	95.4	96.4	93.5	84.4	74.9	(23)
			MCWB	56.8	61.7	65.3	68.2	71.0	74.6	80.2	79.5	76.6	70.8	63.9	58.4	(24)
		10%	DB	67.7	75.6	88.0	100.7	105.8	104.2	96.7	93.6	93.6	91.5	82.2	71.7	(25)
			MCWB	55.6	60.1	64.7	67.3	71.2	75.7	80.2	79.9	77.6	70.6	63.1	57.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.1	68.5	72.1	77.4	81.0	84.8	86.6	85.4	84.4	79.4	69.9	64.4	(27)
			MCDB	73.7	80.4	88.3	95.9	98.9	97.2	92.3	91.9	92.3	89.1	82.2	74.1	(28)
		2%	WB	61.2	66.0	69.8	73.9	79.0	83.1	85.1	84.0	82.3	77.7	68.0	62.0	(29)
			MCDB	68.7	77.4	85.3	94.9	97.6	93.9	91.4	90.2	88.4	86.9	79.9	71.4	(30)
		5%	WB	59.3	64.1	67.9	71.4	77.2	81.7	84.1	83.0	81.3	76.1	66.8	60.1	(31)
			MCDB	66.5	74.6	84.3	94.9	96.6	92.9	90.7	88.9	87.9	85.5	79.1	70.0	(32)
		10%	WB	57.5	62.3	66.5	69.8	75.5	80.4	82.9	82.2	80.3	74.5	65.5	58.4	(33)
			MCDB	64.7	72.2	82.6	92.9	95.7	93.1	89.4	88.0	87.3	85.0	77.5	68.7	(34)
(35)	Mean Daily Temperature Range	MDBR	20.2	22.2	24.3	24.8	22.7	18.2	12.7	12.3	14.9	21.3	23.4	22.2		
		5% DB	MCDBR	24.1	25.3	28.0	27.3	25.3	22.0	16.2	14.9	18.1	23.2	24.8	25.7	(36)
			MCWBR	10.7	9.7	8.0	8.4	8.1	7.0	5.8	5.9	6.1	7.2	8.8	10.9	(37)
		5% WB	MCDBR	19.5	21.4	24.8	23.2	21.7	16.5	13.3	12.3	13.7	17.2	21.3	21.8	(38)
			MCWBR	10.1	9.2	8.3	8.2	8.7	7.0	6.3	5.7	6.0	6.1	8.3	10.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.507	0.482	0.499	0.573	0.670	0.741	0.748	0.653	0.579	0.656	0.630	0.569		
		taud	1.822	1.942	1.931	1.789	1.639	1.555	1.567	1.759	1.878	1.625	1.625	1.703		
		Ebn at Noon	222	246	252	238	215	200	198	216	227	195	185	195		
		Edh at Noon	57	55	59	70	82	89	87	71	62	75	70	62		
(44)	All-Sky Solar Radiation	RadAvg	960	1360	1785	2095	2144	1896	1565	1539	1598	1458	1131	961		
		RadStd	117	110	83	66	94	140	136	130	140	70	75	71		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)		N/A	N/A	+1.89	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page